

METEOR: Evidence-weighted reaction selection reconciles enzyme prediction with growth feasibility (Supplementary material)

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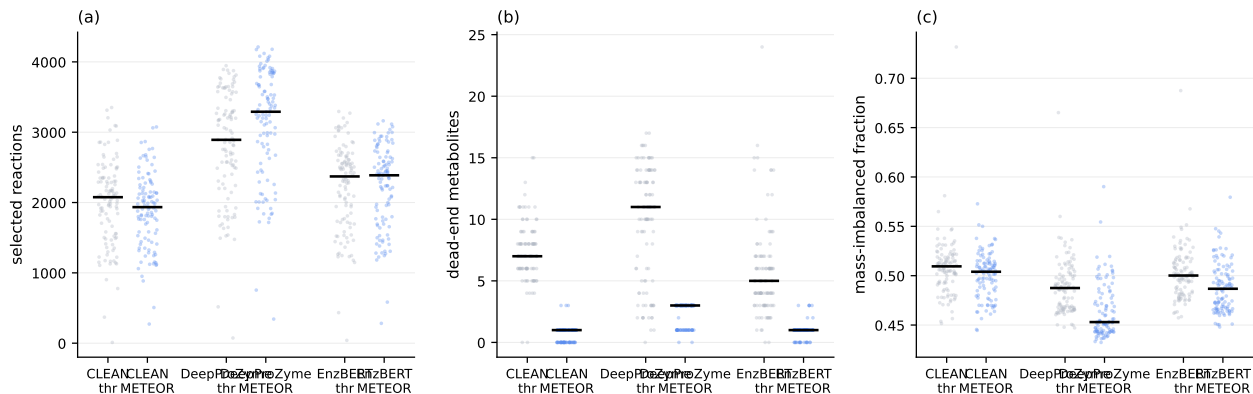


Fig. S1. Per-genome distributions behind Table S25 across the 108-GCF panel: (a) selected reactions, (b) dead-end metabolites, (c) mass-imbalanced fraction, for the $\tau=0.5$ threshold network (grey) and the METEOR network (blue) of each predictor. Each point is one genome; horizontal bars mark medians. Dead-ends and mass imbalance are measured identically on both sides with MEMOTE’s dead-end and mass-balance checks on the COBRA submodel.

S1. Parameters

The scripts and per-genome records behind every analysis in this supplement are listed in the repository README (Availability section of the main text). Figure S1 shows the per-genome mass-imbalance distributions behind Table S25. Default values used throughout the study are: $p = 2$ and $\mu = 3$ (exponent and weight of the evidence-weighted parsimony penalty $\mu(1 - w_j)^p$ added to the log-odds cost; the distinguishing term in Eq. (2) of the main text); $\gamma_{\min} = 0.1$ (lower bound on the selected Gram-specific SEED biomass reaction); $\lambda = 10^{-4}$ (parsimony weight on flux $|v_j|$); $\alpha = 0.4$ (per-EC confidence-update step size in the update equation of main-text Section 2.2); $\epsilon = 10^{-6}$ (log-odds smoothing in the cost equation); $\delta = 0$ (forced-flux floor, disabled; a distinct parameter that would require $|v_j| \geq \delta$ for every selected reaction); decision threshold $\tau = 0.5$ when a hard EC call is required downstream (e.g. for pathway completeness).

Reactions with no EC annotation receive no special treatment. The implementation raises their confidence from zero to a floor of $w_j = 10^{-6}$ before applying the cost equation, giving a cost of approximately 16.1 at these defaults. Apart from fixed medium reactions, this strongly

disfavours their selection. At $w_j = 0.5$, the cost is 0.75; it crosses zero at approximately $w_j = 0.611$.

We did not tune $\gamma_{\min}$, λ or α to maximise any metric on the holdout: all three were fixed during earlier development on *E. coli* and frozen before any holdout evaluation. Sensitivity to these choices is reported in Section S5 of this document.

We classify organisms as Gram-negative or Gram-positive by NCBI taxonomy, which selects both the SEED biomass reaction variant and a set of precomputed tight flux bounds obtained by flux-variability analysis on the full universal model under the defined medium.

S1.1 Complete optimisation protocol

The following steps run between the score matrix and the MILP. We list them so the pipeline can be reproduced end to end.

Input guard. We skip a genome when its baseline prediction matrix has fewer than $0.9\times$ the row count of the EnzBERT matrix for the same assembly. This is the exclusion criterion behind the 108-genome panel.

Score truncation. Each protein keeps its five highest-scoring ECs; all other entries become zero before the noisy-OR of main text Eq. (1). Retaining the full row would let the many small nonzero scores that some predictors assign per protein saturate w_j .

Stoichiometry and bounds. We build $\mathbf{S}$ from the SEED universal with reversible transport reactions expanded, then intersect the universal bounds with the Gram-specific tight bounds above: $lb \leftarrow \max(lb, lb_{\text{tight}})$ and $ub \leftarrow \min(ub, ub_{\text{tight}})$.

Excluded reactions. Reactions that cannot carry flux under those bounds cannot be selected. We also exclude the biomass reaction for the other Gram class; the biomass reaction used as the objective remains available.

Biomass-feasible skeleton. We solve an LP maximising v_{biomass} subject to $\mathbf{S}\mathbf{v} = \mathbf{0}$ and the bounds, then take every reaction with $|v_j| > 10^{-6}$ in that solution as the skeleton. The LP never sees the genome’s scores, so the skeleton is a property of the Gram group and medium rather than of the organism.

Candidate mask. Reaction j is selectable if $w_j \geq w_{\min} = 0.01$, or j lies in the skeleton, or j is a medium exchange; excluded reactions are then removed. Every remaining y_j is fixed to zero, which is what keeps the MILP tractable on a 47,880-reaction universal.

MILP bounds and stopping rules. $\gamma_{\min} = 0.1$ (biomass floor); $\gamma_{\max} = 2.5$ (upper bound on normalised biomass flux); $\lambda = 10^{-4}$; and forced-flux floor $\delta = 0$ (disabled). CBC runs with four threads, a 600 s limit and a 5% relative optimality gap. We record the wall time around the solve call, which includes the solver interface’s model input and output. The interface reports a solve stopped at the time limit with a feasible incumbent as optimal. We therefore report solve calls taking at least 600 s as a wall-time proxy for time-limited runs. The long calls cluster from about 650 s, with none between 600 and 620 s. On the DeepProZyme-vanilla panel 97 of 108 calls took less than 600 s and 11 took at least 600 s; all twelve configurations are in Table S1. Solver logs, incumbent objectives and dual bounds are needed to certify termination reasons and achieved gaps.

The MILP operates under a defined medium: 36 exchange reactions representing essential

nutrients have their uptake bounds set to the universal model’s defaults, while all other exchange reactions are blocked for uptake (lb=0) but may carry secretion flux. The post-solve growth check uses this same medium. MEMOTE and the additional downstream FBA check use a separately inferred uptake medium, as described in Section S7.1.

The verify-and-repair step (main-text Section 2.2) checks whether reactions with $y_j > 0.5$ support $\max v_{\text{biomass}} \geq 0.05$ in a standalone FBA; 5.9% of raw solutions fail this check, and an LP adds minimal-flux support to the biomass floor for these. Every model reported here passed the post-repair check.

Solver protocol and solution status. Solver settings are given above (MILP bounds and stopping rules). We set no random seed, and four-threaded CBC does not guarantee bit-identical re-runs: degenerate optima of equal objective value can differ by a small number of reactions between runs. We measured that spread by re-solving the whole DeepProZyme-vanilla panel a second time under identical settings: 102 of 108 genomes returned the identical selected set, the mean change in selected-set size was 0.0 reactions with a standard deviation of 1.0, and the mean symmetric difference was 1.6 reactions (mean Jaccard 0.998). The differences the paper reports between conditions are 44 to 906 reactions, one to three orders of magnitude above this spread. The released implementation can also select Gurobi or HiGHS when available, but the reported runs used CBC. A feasible incumbent satisfies the numerical formulation: mass balance $\mathbf{S}\mathbf{v} = \mathbf{0}$, inclusion-gated bounds $L_j y_j \leq v_j \leq U_j y_j$, and $v_{\text{biomass}} \geq \gamma_{\min}$. The configured relative-gap target is 5%; elapsed solve time alone is insufficient to certify the achieved gap. All reported selected sets were independently checked for growth after repair.

Candidate-set size. On the DeepProZyme-vanilla panel the candidate mask admits $17,528 \pm 341$ reactions per Gram-negative genome ($n=63$) and $17,569 \pm 208$ per Gram-positive genome ($n=45$) of the 47,880 in the universal. The biomass-feasible skeleton contributes 15,580 (Gram-negative) and 15,578 (Gram-positive) of these; reactions with organism evidence ($w_j \geq 0.01$) number 5,452 on average (range 1,364–6,192), and 12,741 skeleton reactions per genome carry no evidence. The medium adds 36 exchanges, and 6,038 reactions are structurally excluded.

Repair frequency by configuration. Table S1 re-solves the eleven configurations other than DeepProZyme-vanilla at the published settings with the repair count recorded (the original run did not persist it) and adds the DeepProZyme-vanilla rerun of the preceding paragraph. Selected-set sizes match the deposited runs (for example 1,890 CLEAN-vanilla and 2,228 EnzBERT-vanilla against Supplementary Table S25), and the table breaks the main-text aggregate (77 of 1,296, 5.9%) down by configuration.

We re-maximise biomass over each selected set alone (main text, main-text Section 2.2, gives the big- M mechanism that this check targets), under the medium and bounds the MILP used, and repair from the same candidate set where it falls short; three genomes in the deposited DeepProZyme-vanilla solution set carry such a repair, adding 59, 64, and 88 reactions. Every deposited selected set clears the floor on its own after repair, the lowest at 0.38 against $\gamma_{\min}=0.1$. The deposited selections therefore represent the repaired set, whereas the deposited flux vectors and biomass fluxes retain the original MILP witness and should not be read as

Table S1. Post-solve repair by configuration (108 genomes each). Repaired: genomes whose raw MILP solution failed the standalone growth check; added: reactions the repair LP added (median, range); time limit proxy: solve calls taking at least 600 s (Section S1.1); time: median wall time of the solve call.

Predictor	Cutoff	Repaired	Added (median, range)	Limit proxy	Time (s)
CLEAN	vanilla	4	119 (1–230)	83	716
CLEAN	filt30	15	105 (1–195)	95	698
CLEAN	filt50	7	157 (82–226)	97	685
CLEAN	filt70	23	86 (1–224)	93	698
DeepProZyme	vanilla	3	64 (59–88)	11	137
DeepProZyme	filt30	0	–	21	330
DeepProZyme	filt50	0	–	21	304
DeepProZyme	filt70	2	130 (53–207)	21	226
EnzBERT	vanilla	3	152 (73–219)	40	343
EnzBERT	filt30	4	158 (59–222)	55	652
EnzBERT	filt50	5	113 (52–214)	46	335
EnzBERT	filt70	11	111 (1–226)	41	327
All		77/1,296	98 (1–230)	624	

a flux witness for reactions added during repair. Structural results were recomputed from the repaired submodels.

S2. Retraining

To ensure systematic evaluation, we retrain baselines at four sequence-identity cutoffs: 0.30, 0.50, 0.70, and an unfiltered baseline (denoted vanilla, i.e., the original publicly released checkpoint). At a specified cutoff X , we eliminate all proteins that share $\geq X\%$ identity with the 227 holdout proteins from the training pool. We retrain CLEAN on its own filtered split100 data. EnzBERT and DeepProZyme are retrained on SwissProt 2021_04 to ensure parity between those two baselines. All three baselines exclude holdout proteins at each sequence-identity cutoff. Repository URLs for the training splits and code are provided in the Availability section of the main text.

The monotone drop in baseline accuracy from vanilla to filt70 to filt50 to filt30 (e.g. EnzBERT top-1 $41 \rightarrow 38 \rightarrow 35 \rightarrow 22$ on the full $n=125$ SwissProt holdout, before genome-resolution into the 108-GCF panel; on the smaller $n=66$ genome-resolved subset only EnzBERT falls steadily across the cutoffs ($24 \rightarrow 23 \rightarrow 21 \rightarrow 17$), DeepProZyme is flat at 17, and CLEAN varies without a clear direction, which the smaller sample makes noisy; Table S2) is expected: stricter filtering removes more training examples with close similarity to the holdout proteins, leaving the model with fewer examples of the exact function families that appear in the holdout. The large step at the 30% cutoff (filt50 $35 \rightarrow$ filt30 22 for EnzBERT) is consistent with the 30%–50% band containing a disproportionate share of the nearest-neighbour training exemplars for holdout proteins.

S2.1 Filtered-retraining holdout-66 results

Table S2 reports protein-level fidelity on holdout-66 across all four sequence-identity retraining cutoffs; the vanilla rows are summarised in main-text Section 3.4. filt30/filt50/filt70 denote baselines retrained after removing training proteins sharing $\geq 30/50/70\%$ identity with the 227 holdout proteins. Under the net-only comparison (baseline and METEOR mapped through the identical EC-vocabulary normalisation), top-1 is exactly preserved in all 12 configurations. METEOR changes no top-1 call anywhere, so the paired test has no discordant pair to work with and we report none, while $F_{\max}$ stays within ± 0.012 with no consistent sign. In this holdout, the update therefore changes no top-1 call across predictors and retraining cutoffs. The small positive top-1/ $F_{\max}$ gains reported by a naive (non-net-only) comparison are an EC-vocabulary artefact (Section S8).

Table S2. Protein-level fidelity on holdout-66 across all baseline $\times$ retraining-cutoff configurations, net-only (baseline and METEOR mapped through the identical EC-vocabulary normalisation). B = baseline; $+M$ = METEOR-refined.

	top-1 $\uparrow$		$F_{\max}$ $\uparrow$		$\Delta F_{\max}$
	B	$+M$	B	$+M$	
Baseline					
CLEAN-vanilla	26	26	0.429	0.428	± 0.000
CLEAN-filt30	22	22	0.364	0.364	± 0.000
CLEAN-filt50	25	25	0.421	0.421	± 0.000
CLEAN-filt70	28	28	0.453	0.453	± 0.000
DeepProZyme-vanilla	17	17	0.308	0.307	-0.001
DeepProZyme-filt30	17	17	0.318	0.317	± 0.000
DeepProZyme-filt50	17	17	0.318	0.319	± 0.000
DeepProZyme-filt70	17	17	0.314	0.314	± 0.000
EnzBERT-vanilla	24	24	0.385	0.385	± 0.000
EnzBERT-filt30	17	17	0.292	0.288	-0.004
EnzBERT-filt50	21	21	0.340	0.340	$+0.001$
EnzBERT-filt70	23	23	0.376	0.365	-0.012

Note. $n=66$ proteins. Top-1 is identical in every configuration, so no protein changes status and there is nothing for a paired test to compare. $\Delta F_{\max}$ lies within $[-0.012, +0.001]$, without a consistent sign.

S2.2 Calibration analysis

Table S3 reports calibration metrics on the same holdout-66 set, under the evidence-weighted cost used throughout the paper. We report it to document that the monotone per-EC update does not target calibration, and that where it changes calibration it does so unevenly across predictors.

Calibration changes are predictor-specific rather than systematic. ECE improves in 6 of 12 configurations and Brier in 8. DeepProZyme, whose initial miscalibration is the highest

Table S3. Calibration on holdout-66 ($n=66$). B = baseline; $+M$ = METEOR-refined. Bold indicates improvement. ECE (15 bins) improves in 6 of 12 configurations and Brier in 8.

Configuration	ECE ↓		Brier ↓	
	B	$+M$	B	$+M$
CLEAN-vanilla	0.140	0.151	0.129	0.136
CLEAN-filt30	0.301	0.299	0.279	0.278
CLEAN-filt50	0.319	0.309	0.275	0.274
CLEAN-filt70	0.333	0.334	0.307	0.309
DeepProZyme-vanilla	0.436	0.427	0.358	0.344
DeepProZyme-filt30	0.326	0.271	0.274	0.248
DeepProZyme-filt50	0.332	0.290	0.289	0.266
DeepProZyme-filt70	0.340	0.296	0.286	0.261
EnzBERT-vanilla	0.283	0.295	0.257	0.253
EnzBERT-filt30	0.130	0.158	0.143	0.154
EnzBERT-filt50	0.199	0.200	0.199	0.196
EnzBERT-filt70	0.305	0.339	0.291	0.299

(ECE ≈ 0.3 – 0.4), improves on both metrics at all four cutoffs. CLEAN improves on both at filt30 and filt50 and worsens on both at vanilla and filt70. EnzBERT worsens on ECE at all four cutoffs while its Brier score improves at two. The confidence update is monotone within each EC column and does not target calibration, so it can move the two metrics in opposite directions, as it does for EnzBERT. We make no calibration claim.

S3. Ablation and controls

We evaluate METEOR’s components by ablation on the 108-GCF panel, DeepProZyme-vanilla, using the fair net-only protein comparison and the pathway-detection metric of main-text Section 3.2. All conditions use the hard-biomass MILP unless stated otherwise.

Conditions. B0: baseline only (no refinement). B2: full MILP with the biomass floor removed ($\gamma_{\min}=0$). B3: full METEOR (evidence-weighted penalty, hereafter evw; $p=2$, $\mu=3$). uniform ($p=0$): the same hard-biomass MILP with a confidence-independent parsimony penalty ($\mu(1 - w_j)^0$), isolating the contribution of evidence weighting. B2 and B3 share the same noisy-OR confidence w_j and log-odds cost; B0 supplies the same baseline scores but does not solve the MILP. B2 and B3 differ only in the biomass floor; B3 and uniform differ only in the penalty exponent p . The strongest control on the sequence-derived cost is the internal deletion-recovery experiment under evidence-derived, uniform, and shuffled costs (main-text Section 3.3), which quantifies off-reference selection while holding the feasible candidate set fixed.

Ablation findings. Detection counts in Table S5 are not size-matched (Section S3.1) and compare conditions with one another; they are not a pathway-inference claim. At the protein level (Table S4, net-only), B0, B2, and B3 have identical top-1 (17/66), while $F_{\max}$ changes from

Table S4. Protein-level ablation on the holdout-66 evaluation set ($n=66$, DeepProZyme-vanilla, net-only).

Condition	top-1 (/66) $\uparrow$	$F_{\max}$
B0 (baseline only)	17	0.308
B2 (no biomass floor)	17	0.307
B3 (full METEOR)	17	0.307
uniform ($p=0$)	16	0.305

Note. Net-only. B2/B3 preserve baseline top-1; the confidence-independent uniform penalty drops one call.

Table S5. System-level ablation: pathway detection ($P=151$ SEED-representable pathways) on the 108-GCF panel, DeepProZyme-vanilla. Detected = mean pathways/genome with ≥ 1 EC / $\geq 50\%$ of ECs present ($B \rightarrow M$). sub-thr. EC: fraction of available nonzero sub-threshold ECs ($0 < \text{score} < 0.5$) that the condition selects.

Condition	≥ 1 EC	$\geq 50\%$	sub-thr. EC
B0 (baseline only)	123.9 (–)	16.6 (–)	–
B2 (no biomass floor)	123.9 $\rightarrow$ 132.2	16.6 $\rightarrow$ 35.3	18.6%
B3 (full METEOR)	123.9 $\rightarrow$ 132.5	16.6 $\rightarrow$ 36.0	19.1%
uniform ($p=0$)	123.9 $\rightarrow$ 121.9	16.6 $\rightarrow$ 22.6	10.8%

Note. evw (B3) detects +8.6 pathways at ≥ 1 EC; the confidence-independent uniform penalty loses 2, pruning more nonzero signal.

Table S6. Structural properties of each condition’s MILP selected set, 108 genomes, DeepProZyme-vanilla. n_{selected} : reactions selected by the MILP ($y > 0.5$). dead ends/MI frac: per COBRA submodel (fraction failing elemental balance). Growing: nonzero FBA biomass under the defined medium.

Condition	n_{selected}	dead ends $\downarrow$	MI frac $\downarrow$	Growing $\uparrow$
B2 (no biomass floor)	3,077	–	–	0/108
B3 (full METEOR, evw)	3,121	2.35	0.464	108/108
uniform ($p=0$)	2,215	1.74	0.492	108/108

Note. B2 vs. B3 (near-equal size, 0/108 vs. 108/108 growing) isolates the biomass floor. evw (B3) retains more reactions than uniform (3,121 vs. 2,215); uniform pruning does not use reaction confidence. B2’s non-viable models are not MEMOTE-profiled (dead ends/MI n/a).

0.308 to 0.307. The uniform penalty is the exception: it drops one top-1 call (16/66).

At the system level (Tables S5 and S6), the two penalties produce different selected sets. evw (B3) detects +8.6 pathways per genome at ≥ 1 EC and roughly doubles the $\geq 50\%$ count, whereas the uniform penalty loses two pathways at ≥ 1 EC and adopts only 10.8% of the available nonzero sub-threshold ECs versus evw’s 19.1%: confidence-independent pruning discards nonzero sub-threshold evidence that evw retains. The biomass floor (B2 vs. B3) barely changes detection, so the difference between these conditions is associated with the evidence-weighted

penalty rather than the biomass floor.

The structural table isolates each component. B2 and B3 are near-equal in size (3,077 vs. 3,121 selected reactions) yet zero of 108 B2 models grow, versus 108/108 for B3: the biomass floor requires a biomass-supporting route, making the selected submodels growth-feasible without substantially enlarging them. evw (B3) also retains more reactions than the uniform penalty (3,121 vs. 2,215). This comparison concerns reaction retention, not isozymes, because the formulation contains no protein-level variables.

Candidate-mask ablation. The ablations above hold the candidate mask fixed. Table S7 varies it on the same 108 genomes (DeepProZyme-vanilla, published settings, same solver settings), with the published mask re-solved in the same batch as a control. No skeleton keeps evidence ($w_j \geq 0.01$) and medium reactions only: the MILP is infeasible in every genome, so evidence alone never contains a growth-supporting route. Skeleton only makes no evidence reaction selectable: the MILP returns a minimum-cost subset of the scaffold that grows after repair in every genome, with no dead-end metabolites, although every raw solution failed the standalone growth check and needed the repair LP (mean 69 reactions added), and 106 of 108 solves stopped at the time limit (11 for the full mask). The full mask selects about 1,100 more reactions than the scaffold alone and incurs 2.4 dead ends. Growth feasibility and the absence of dead ends are therefore properties the scaffold supplies; the evidence-weighted cost decides which reactions join it.

Table S7. Candidate-mask ablation, 108 genomes, DeepProZyme-vanilla. Metrics as in Table S6; repaired: genomes whose raw solution needed the repair LP.

Candidate mask	Infeasible	n_{selected}	dead ends ↓	MI frac	Growing	Repaired
full (published, re-solved)	0/108	3,121	2.36	0.464	108/108	2
no skeleton (evidence $\vee$ medium)	108/108	–	–	–	0/108	–
skeleton only (skeleton $\vee$ medium)	0/108	1,985	0.00	0.495	108/108	108

Sequence-cost control: internal deletion-recovery. The deletion-recovery experiment (main-text Section 3.3) tests whether the sequence-derived cost changes selection when the feasible candidate set is held fixed. Associated EC-score columns are ablated, and the MILP is resolved under the evidence-derived cost, a confidence-independent uniform cost, and shuffled costs. The evidence-derived cost selects $4\text{--}58\times$ fewer off-reference reactions than the uniform or shuffled alternatives (Wilcoxon $p < 10^{-18}$, 108 genomes), relative to METEOR’s unperturbed selected set. Because the reference is internal, this result measures stability rather than biological accuracy.

S3.1 Pathway coverage: the size-matched control

Set size confounds any pathway comparison between METEOR and a hard-threshold baseline, because METEOR’s MILP-selected EC set is larger than the baseline’s (≥ 0.5) set. Matching the sizes removes the confound directly. We therefore score METEOR’s selected EC set of size

K against the top- K SEED-representable ECs by maximum baseline score, on the same 151 pathways and the same 108 genomes (Table S8).

The result is negative (summarised in main-text Section 3.2). Against the fixed-threshold baseline, METEOR gains 4.7–9.1 percentage points of mean pathway coverage. Against the size-matched baseline it loses 2.7–2.8 points, and falls below it in 107 of 108 genomes for CLEAN and in all 108 for DeepProZyme and EnzBERT. The apparent gain therefore reflects how many ECs METEOR returns, not better pathway recovery, and we make no pathway-inference claim in the main text.

Table S8. Size-matched pathway coverage on the 108-GCF panel (151 SEED-representable pathways). thr = baseline at score ≥ 0.5 ; top- K = size-matched baseline, the K highest-scoring SEED-representable ECs with K = METEOR’s selected EC count. Values are mean fraction of pathway ECs present. Δ columns are percentage points. lower in counts genomes where METEOR falls below the size-matched baseline.

Baseline	thr	top- K	METEOR	Δ vs. thr	Δ vs. top- K	lower in
CLEAN	0.168	0.242	0.215	+4.73	−2.69	107/108
DeepProZyme	0.214	0.332	0.305	+9.10	−2.70	108/108
EnzBERT	0.177	0.262	0.234	+5.74	−2.79	108/108

For completeness we also record thresholded detection counts at three stringencies (Table S9). These favour METEOR, but they do not rescue a pathway claim: the least stringent detection measure is insensitive to how much of a pathway is present. One observation from that table does survive independently of the comparison baseline: 88–99% of the pathways METEOR detects but the threshold baseline misses are associated with nonzero sub-threshold evidence ($0 < \text{score} < 0.5$) rather than zero-evidence reactions.

The additional selection is limited rather than indiscriminate: of the ECs with nonzero sub-threshold evidence ($0 < \text{score} < 0.5$) available per genome, METEOR selects only 308/4146 (7.4%, CLEAN), 328/1715 (19.1%, DeepProZyme), and 334/3711 (9.0%, EnzBERT); and of the pathways that carry recoverable sub-threshold evidence (baseline-missed but with ≥ 1 such EC), it recovers only 19.3% (8.4/43.4, CLEAN), 39.3% (9.5/24.1, DeepProZyme), and 21.7% (7.7/35.6, EnzBERT). METEOR therefore does not admit every nonzero-scored EC; the evw cost prunes the large majority of that evidence and the network selects a small, biomass-associated subset.

S4. Alternative repair

To test whether simpler repair methods can recover the structural quality that METEOR achieves, we compared two alternatives on the same 108-GCF panel under the same defined growth medium used by METEOR (29 SEED compounds plus 7 essential inorganics; 36 importable exchanges total).

Table S9. Pathway detection counts (mean pathways/genome of 151) at three stringencies. A pathway is detected if the fraction of its ECs present is $\geq 1/n$ (≥ 1 EC), ≥ 0.25 , or ≥ 0.50 . The ≥ 5 -EC cutoff behind the 151 is what keeps these three apart: a pathway with four or fewer SEED-representable ECs reaches ≥ 0.25 on a single EC, so that column would repeat the ≥ 1 EC one. Relaxing the cutoff to ≥ 3 or ≥ 1 admits 162 and 166 pathways and changes no conclusion. The ≥ 1 EC gain runs +5.9 to +9.7 across all three settings, the $\geq 50\%$ ratio 1.79–2.17 $\times$, and the nonzero sub-threshold share of METEOR-only detections stays at 88–99%. M-only weak/gap: of pathways METEOR detects (≥ 1 EC) that the baseline misses, the split between nonzero sub-threshold driving ECs ($0 < \text{score} < 0.5$) and zero-evidence ECs.

Baseline	≥ 1 EC $B \rightarrow M$	$\geq 25\%$ $B \rightarrow M$	$\geq 50\%$ $B \rightarrow M$	M-only (weak/gap)
CLEAN	107.6→114.1	42.0→55.8	10.2→20.0	8.3/0.1
DeepProZyme	123.9→132.5	56.3→80.2	16.6→36.0	9.1/1.3
EnzBERT	115.4→121.5	43.7→61.3	10.4→22.1	7.6/0.1

~~Greedy topology-only repair. A greedy topology-only alternative to the LP repair (iteratively adding the lowest-log-odds-cost unselected reaction touching a dead-end metabolite, capped at 20,000 iterations, starting from the DeepProZyme-vanilla reaction-level threshold network, dead-end fraction 0.789) adds a mean of 16,445 reactions yet plateaus at dead-end fraction 0.235, supports biomass growth in 0 of 108 genomes (all hit the iteration cap), and takes a median of 64 min per genome: it does not reach the LP repair’s quality.~~

Comparison against a general-purpose gap-filler. An earlier implementation omitted the medium from the thresholded draft, which has no uptake reactions of its own. We discarded its invalid zero-growth output and carried the shared medium into both arms of the corrected comparison below.

The corrected comparison fixes everything except the algorithm. Both arms start from the same DeepProZyme per-EC threshold draft (median 2,892 reactions, the same construction as main-text Figure 2a/Table S25), solve under the same medium, and draw from one shared candidate set R (median 20,156 of the universal’s 47,880). We build R without consulting METEOR’s confidences, using EC-annotated reactions, the biomass-feasible skeleton, and the medium exchanges, less those structurally excluded. The candidate set therefore does not use METEOR’s reaction-confidence costs.

The formulations differ, and that is the point. METEOR repairs with a linear program minimising flux through non-core reactions; COBRApy’s¹ gap-filler solves a mixed-integer program minimising the number of reactions added, one binary variable per candidate. Across 108 genomes the LP returned a growing solution every time, in a median 9.6 s, adding a median of 82 reactions. The MILP returned no solution within a 3,120 s budget in any genome, about 330 $\times$ the LP’s median, and was faster in no genome.

This does not support two readings. The MILP is not shown to be unsolvable here: it was cut off and never proved infeasible, so its runtime and the runtime ratio are both lower bounds. Neither is the LP shown to be the better algorithm, since it solves an easier problem; on a 95-reaction textbook model, where both finish in 0.03s, the two return the same minimal set of four reactions. The finding concerns scale alone. Within this time budget, minimum-count gap-filling does not reach a solution on a candidate set this size, which is why METEOR repairs with a flux-parsimony LP. The per-genome records behind this comparison are not deposited, so it cannot be re-read from the deposited records; the scripts that produce and aggregate them are in the release.

Holding the predictor fixed gives a sharper comparison still: Section S7.4 gap-fills the same DeepProZyme threshold draft with the parsimony repair METEOR uses internally, and those models grow. That is the arm against which evidence-weighted selection shows no accuracy advantage (EC-level recall 0.784 versus 0.767). The deletion-recovery experiment includes a separate two-stage control constructed on its fixed candidate set.

S5. Parameter sensitivity

We assess sensitivity to the three hyperparameters that most directly shape METEOR’s output: the score-truncation depth k (input), the confidence-update strength α (per-protein), and the evidence-weighted-parsimony penalty (p, μ) (network). We fixed the remaining hyperparameters ($\gamma_{\max} = 2.5$, $\lambda = 10^{-4}$, $\epsilon = 10^{-6}$, $\delta = 0$, aggregation = noisy-OR) at their production values and froze them before holdout evaluation; $\gamma_{\min}$ is treated separately below. A full Cartesian sweep of every parameter was not performed; we report the three available sweeps and the frozen defaults for the remaining parameters.

Score-truncation depth k . Each protein keeps its k highest-scoring ECs before the noisy-OR (main-text Section 2.2). We swept $k \in \{1, 3, 5, 10\}$ and no truncation across 53 genomes with the log-odds cost alone, without the evidence-weighted parsimony term $\mu(1 - w_j)^p$. The sweep code is released, but the per-genome records are not, so the table documents an earlier sensitivity analysis rather than a reproducible analysis of the final formulation.

Table S10. Sensitivity to the truncation depth k . Protein-level metrics are computed on the legacy 67-protein holdout; selected reactions is the mean over 53 genomes. This sweep was run with the log-odds cost alone, so the reaction counts are larger than the evidence-weighted counts reported in the main text.

k	top-1	top-3	$F_{\max}$	selected reactions
1	18/67	23	0.317	4,819
3	18/67	23	0.317	5,932
5 (default)	18/67	23	0.317	6,596
10	18/67	23	0.317	7,509
none	18/67	23	0.318	9,607

The protein-level output does not move at all (Table S10): top-1 and top-3 are identical at every k , and $F_{\max}$ shifts by 0.001 only when truncation is removed entirely. The selected reaction count, in contrast, rises monotonically and roughly doubles between $k=1$ and no truncation. We therefore describe k as a parsimony control rather than an accuracy control, and we do not claim the method is insensitive to it: k decides how large the reconstructed model is in this earlier formulation. The default $k=5$ was fixed before holdout evaluation; this sweep does not identify an accuracy optimum.

Confidence-update strength α . We swept $\alpha \in \{0.2, 0.4, 0.6\}$ (equivalently $\beta = \alpha/0.4 \in \{0.5, 1.0, 1.5\}$) on the 66 net-only-evaluable holdout proteins (DeepProZyme baseline, production $p=2, \mu=3$ MILP). α rescales the posterior update only; the MILP solution is identical across α , so both the baseline and METEOR scores are mapped through the same EC-vocabulary normalisation (net-only) as elsewhere.

Table S11. Sensitivity of net-only per-protein metrics to the confidence-update strength α (equivalently $\beta = \alpha/0.4$) on $n=66$ holdout proteins (hard-biomass MILP, DeepProZyme baseline, production $p=2, \mu=3$). $\alpha=0.4$ is the production default.

α (β)	n	top-1 $\uparrow$	top-3 $\uparrow$	$F_{\max}$ $\uparrow$
0.2 (0.5)	66	17	22	0.308
0.4 (1.0)	66	17	22	0.307
0.6 (1.5)	66	17	22	0.302

Top-1 and top-3 are exactly invariant across the range (Table S11; the per-EC update never crosses a first-to-second-EC gap); $F_{\max}$ varies by ≤ 0.006 and is highest at the most conservative $\alpha=0.2$. The production default $\alpha=0.4$ was frozen before holdout evaluation and is not the $F_{\max}$ -optimal value, consistent with the statement that it was not tuned to the holdout. Calibration metrics under the same update are reported separately in Table S3.

Parsimony penalty (p, μ) . The exponent p and weight μ of the penalty $\mu(1 - w_j)^p$ govern how aggressively low-confidence reactions are pruned, so their natural readout is network-level. We evaluate the full $p \in \{0, 1, 2, 3\} \times \mu \in \{1, 3, 5, 8\}$ grid against reaction-level references from the six curated GEMs, the only cohort with curated reaction sets, reporting MetaNetX-bridged² precision, recall, and Jaccard of the repaired selected set against the curated model, plus mean network size.

Reaction overlap with the curated models is nearly flat across the entire grid (Table S12; Jaccard 0.12 ± 0.01): larger μ , or the confidence-independent $p=0$, prunes more aggressively and trades recall for precision at roughly constant Jaccard. The production default ($p=2, \mu=3$) is not the Jaccard- or precision-optimal cell; uniform $p=0, \mu=5$ attains the highest Jaccard. The default was therefore not chosen to maximise either displayed metric, and evidence weighting

Table S12. Sensitivity of curated-GEM reaction recovery to the parsimony penalty (p, μ), mean over the six curated-GEM model organisms (DeepProZyme baseline; $p=0$ is the confidence-independent uniform penalty). ($p=2, \mu=3$) is the production default.

p	μ	Prec.	Recall	Jaccard	n_{selected}
0	1	0.173	0.286	0.120	3188
0	3	0.200	0.266	0.128	2580
0	5	0.222	0.237	0.129	2116
0	8	0.243	0.185	0.117	1532
1	1	0.165	0.290	0.116	3378
1	3	0.171	0.287	0.119	3222
1	5	0.178	0.284	0.122	3084
1	8	0.181	0.281	0.123	2994
2	1	0.163	0.292	0.116	3439
2	3	0.166	0.290	0.117	3360
2	5	0.168	0.289	0.118	3305
2	8	0.171	0.288	0.119	3249
3	1	0.162	0.293	0.116	3472
3	3	0.163	0.292	0.116	3418
3	5	0.165	0.291	0.117	3387
3	8	0.166	0.290	0.117	3354

Note. $n=6$ organisms; the production default is $p=2, \mu=3$.
The raw sweep output is not included in the release.

does not improve static curated-model overlap on this grid. Its measured effect appears instead in the internal deletion-recovery comparison (in-text Section 3.3). Keeping a redundant route that the evidence supports adds a reaction to the curated reference that is not listed, so higher overlap is not what the penalty targets.

Biomass floor $\gamma_{\min}$. The remaining MILP parameter of interest is $\gamma_{\min}$, the hard lower bound on biomass flux. The ablation in Section S3 provides a direct comparison: $\gamma_{\min}=0$ (condition B2) vs. $\gamma_{\min}=0.1$ (condition B3, the production default). At the protein level, both conditions yield 17/66 top-1 and $F_{\max} = 0.307$. Their pathway-detection counts are also close (132.2 vs. 132.5 pathways with at least one EC; 35.3 vs. 36.0 at least half complete). The structural result differs: B2 produces 0/108 growth-feasible submodels, B3 produces 108/108. This ablation shows that the production biomass floor supplies feasibility; it does not test sensitivity among positive values of $\gamma_{\min}$. We fixed $\gamma_{\min}=0.1$ during earlier development and froze it before holdout evaluation. Because the SEED biomass reaction is normalised, $\gamma_{\min}=0.1$ is a minimal viability floor, not a quantitative growth-rate prediction.

S6. Additional baselines

The main text’s pathway-detection analysis (main-text Section 3.2) and the MEMOTE structural quality profiles (Section S7) cover the three baselines with full 108-GCF-panel predictions (CLEAN, DeepProZyme, EnzBERT). Three additional baselines (GraphEC,³ a structure-aware graph neural network; MAPred;⁴ and TopEC⁵) are evaluated on the smaller 22-genome Price-149 panel ($n=149$ proteins with experimentally verified EC annotations across 22 organisms). Price et al.⁶ determined those annotations experimentally; ProteInfer⁷ first assembled them as a benchmark, on the grounds that automated methods had labelled these sequences incorrectly or inconsistently in databases such as KEGG. Table S13 reports per-protein fidelity; Table S15 reports pathway completeness. Organism-scale predictions for these three baselines exist only on this smaller panel.

Table S13. Per-protein fidelity on the Price-149 benchmark ($n=149$, 22 genomes, hard-biomass MILP), net-only (baseline and METEOR mapped through the identical EC-vocabulary normalisation). B = baseline; $+M$ = METEOR-refined. All six baseline architectures are evaluated under the same MILP configuration. We ran each predictor with the authors’ released code and default settings, without retraining. The two structure-aware methods took ESMFold-predicted structures rather than the experimental or AlphaFold2 structures their published benchmarks use: GraphEC through the released five-fold ensemble, TopEC through its released structure-based checkpoint at 703 classes. Support also differs by three orders of magnitude across these predictors (Table S14): TopEC scores one EC per protein while GraphEC spreads mass over 5,071, so top-1 and $F_{\max}$ are not comparable across rows. None of these values is the tool’s published benchmark performance.

Baseline	top-1 $\uparrow$		$F_{\max}$	
	B	$+M$	B	$+M$
CLEAN	60	61	0.559	0.539
DeepProZyme	49	49	0.558	0.529
GraphEC	18	18	0.264	0.233
MAPred	50	52	0.550	0.537
TopEC	5	5	0.034	0.034
EnzBERT	53	53	0.477	0.463

Under the net-only comparison, METEOR leaves top-1 nearly unchanged (CLEAN +1, MAPred +2, the rest unchanged; no top-1 regression) but does not improve $F_{\max}$: $\Delta F_{\max}$ ranges from -0.031 (GraphEC) to 0 (TopEC). The earlier apparent gains (e.g. GraphEC $+0.030$) were an EC-vocabulary artefact of a non-net-only comparison. The two largest $F_{\max}$ declines occur for GraphEC and DeepProZyme, for which thousands of ECs per protein are muted (Table S14). TopEC, which scores essentially one EC per protein, is unchanged; CLEAN is an exception to a simple support-size explanation because only 37 of its many nonzero ECs are muted. Top-1 changes only for CLEAN and MAPred. We therefore do not claim METEOR improves out-of-distribution protein-level fidelity; on this benchmark the network-selection step leaves top-1 nearly unchanged while imposing a small, predictor-dependent $F_{\max}$ cost.

Table S14. Output-distribution profile of the six predictors on Price-149 and the resulting net-only $\Delta F_{\max}$. Support = mean number of ECs scored > 0 per protein; conf. = fraction of proteins whose top EC scores ≥ 0.5 ; muted = mean ECs demoted by METEOR per protein. These counts contextualise the observed change in $F_{\max}$.

Predictor	Support	conf.	muted/prot	$\Delta F_{\max}$
GraphEC	5,071	0.34	5,068	-0.031
CLEAN	5,236	0.59	37	-0.020
DeepProZyme	2,715	0.73	2,674	-0.029
EnzBERT	100	0.58	97	-0.014
MAPred	27	0.93	22	-0.013
TopEC	1	0.74	0.07	0.000

We repeat the pathway analysis of main-text Section 3.2 for the three additional predictors on the 22 Price-149 genomes, using the same $P=151$ SEED-representable pathways and reporting the same detection-and-matched-completeness metrics rather than raw average completeness (which is confounded by reaction-set size; main-text Section 3.2, Section S3).

Table S15. Price-149 pathway detection for GraphEC, MAPred, and TopEC (22 genomes, 151 pathways). Detection: mean pathways/genome with ≥ 1 EC / $\geq 50\%$ of ECs present ($B \rightarrow M$). M-only (weak/gap): pathways METEOR detects (≥ 1 EC) that the baseline misses, split into nonzero sub-threshold ($0 < \text{score} < 0.5$) vs. zero-evidence driving ECs.

Baseline	det. ≥ 1 EC $B \rightarrow M$	det. $\geq 50\%$ $B \rightarrow M$	M-only (weak/gap)
GraphEC	75.1 $\rightarrow$ 102.6	4.3 $\rightarrow$ 12.5	29.4 (29.0/0.4)
MAPred	134.8 $\rightarrow$ 134.4	46.8 $\rightarrow$ 57.7	3.0 (2.7/0.3)
TopEC	107.6 $\rightarrow$ 112.8	1.7 $\rightarrow$ 8.4	6.4 (1.2/5.1)

The descriptive pattern matches the main panel (main-text Section 3.2): pathway detection at $\geq 50\%$ completeness increases for all three, and at ≥ 1 EC for GraphEC and TopEC, most strongly for the diffuse structure-aware GraphEC (+27.5 pathways at ≥ 1 EC, $\sim 3\times$ at $\geq 50\%$); MAPred edges down at ≥ 1 EC (134.8 $\rightarrow$ 134.4) while its $\geq 50\%$ count rises (46.8 $\rightarrow$ 57.7). Nonzero sub-threshold evidence drives 98.6% of GraphEC’s METEOR-only detections (29.0/29.4) and 90% of MAPred’s. TopEC is an exception: it commits to essentially one EC per protein (Table S14), so it carries almost no sub-threshold signal to select; 80% of its METEOR-only detections (5.1/6.4) come from zero-evidence gap-fill reactions. For predictors with graded scores, most additional detections are associated with nonzero sub-threshold evidence; for TopEC, they predominantly reflect gap-filling. As in the main analysis, these detection counts do not establish pathway recovery.

S7. Structural quality, external reconstruction context, and computational cost

S7.1 MEMOTE structural quality and comparison with CarveMe

Table S16 gives the EC-level comparison summarised in main text Section 3.4. Full: all ECs METEOR emits from the 6,598-EC ModelSEED mapping; Reference: both methods restricted to the 1,234-EC BiGG-derived vocabulary of the curated models.

Table S16. EC-level comparison against six curated BiGG GEMs, averaged over organisms. S.d. across organisms is at most 0.09 on ratio metrics and 174 on size.

	ECs	Precision	Recall	F1	Jaccard
METEOR, full vocabulary	1,521	0.438	0.884	0.585	0.415
METEOR, reference vocabulary	876	0.761	0.884	0.817	0.694
CarveMe	818	0.822	0.893	0.854	0.746

We compare METEOR and CarveMe⁸ on MEMOTE structural quality under their respective reconstruction settings. The two tools differ in every design dimension: input (bounded EC scores vs. genome), template (SEED universal vs. curated BiGG), namespace, and objective. This is context rather than a controlled comparison. METEOR’s mean MEMOTE total score is higher in these records for all three baselines in the vanilla configurations (0.765–0.788 vs. 0.735–0.739 for CarveMe; 0.765–0.796 across all twelve). METEOR has a higher unbounded-flux fraction (mean 0.75–0.94 in the vanilla configurations, 0.73–0.94 across all twelve, vs. 0.545; lower is better), a metric that favours CarveMe’s smaller, manually curated BiGG template.

Every reported METEOR selected set attains the biomass floor after post-solve verification and repair (108/108; main-text Section 3.1). A downstream FBA on the MEMOTE submodel uses an uptake medium inferred post hoc, which differs from the MILP medium (Section S1). That check runs on a separately constructed submodel, exported to SBML and re-imported before MEMOTE, so it is not directly comparable with the main-text Section 3.1 procedure; it grows in 105–108/108 genomes in the three vanilla configurations (108/108 for DeepProZyme and EnzBERT; 105/108 for CLEAN). Across all twelve configurations the range is 83–108/108, the three lowest being CLEAN-filt70 (83/108), CLEAN-filt30 (94/108)

and CLEAN-filt50 (103/108). The difference concerns the downstream medium and submodel conversion; it does not overturn the fixed-medium post-solve check.

Neither tool’s structural scores should be interpreted as protein-level accuracy: CarveMe over-predicts amino-acid auxotrophies,⁹ and automated pipelines including CLEAN reach only 50–69% accuracy on experimentally verified enzymes.¹⁰ MEMOTE evaluates network-level structural quality, not whether each predicted EC is correct. Namespace-neutral EC-level recall against curated GEMs, the comparison we do rely on in the main text, is reported in Sections S7.4 and S7.5.

S7.2 MEMOTE sub-score breakdown

Table S17 reports the per-section MEMOTE scores across all 12 baseline×cutoff configurations on the 108-GCF panel. Consistency (weight 3) is the only sub-score that varies meaningfully across configurations (84.5–88.1%), driven by the number and composition of reactions selected by the MILP. The four annotation sub-scores are nearly uniform because they derive from the shared SEED universal template. CLEAN-vanilla shows an anomalous gene-annotation score (27.8% vs. 40.0% elsewhere) due to 33 genomes with zero gene annotations in the COBRA submodel; this artefact does not affect the consistency or total-score ranking. The two rightmost columns report, per configuration, the fraction of the 108 genomes with nonzero FBA-optimal biomass flux under the downstream medium inferred post hoc (Growing) and the mean MEMOTE unbounded-flux metric (Unbounded, fraction of reactions carrying unbounded flux in the default condition; lower is better); both are extracted from the same per-genome MEMOTE snapshot reports underlying the sub-scores at left. Growing ranges 76.9–100.0% ($\geq 97\%$ in 9 of 12 configurations); Unbounded ranges 0.73–0.94, versus CarveMe’s 0.545 on the same 108-genome panel (Section S7.1).

Table S17. MEMOTE quality sub-scores for reconstructed models across the 108-GCF panel ($n=108$ genomes per configuration). Consist.–Total values are mean $\pm$ s.d. (%); MEMOTE section weights shown in parentheses; Total is the weighted aggregate. Growing: fraction of genomes with nonzero FBA-optimal biomass flux under the downstream medium inferred post hoc. Unbounded: mean fraction of reactions carrying unbounded flux (MEMOTE consistency test, lower is better). Bold marks the highest value in the Consistency and Total columns.

Baseline	Cutoff	Consist. ($w=3$)	Ann. Met. ($w=1$)	Ann. Rxn. ($w=1$)	Ann. Gene ($w=1$)	Ann. SBO ($w=2$)	Total	Growing (%)	Unbounded
CLEAN	vanilla	87.5 $\pm$ 1.5	68.5 $\pm$ 0.1	65.1 $\pm$ 0.3	27.8 $\pm$ 18.4	76.2 $\pm$ 8.4	76.5 $\pm$ 4.5	97.2	0.75
	filt30	88.1 $\pm$ 2.0	68.5 $\pm$ 0.1	65.1 $\pm$ 0.2	39.6 $\pm$ 3.8	81.6 $\pm$ 1.7	79.6 $\pm$ 1.1	87.0	0.73
	filt50	88.0 $\pm$ 1.7	68.5 $\pm$ 0.2	65.1 $\pm$ 0.7	40.0 $\pm$ 0.0	81.7 $\pm$ 0.9	79.6 $\pm$ 0.8	95.4	0.73
	filt70	86.8 $\pm$ 1.8	68.5 $\pm$ 0.1	65.2 $\pm$ 0.2	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	79.2 $\pm$ 0.7	76.9	0.81
DeepProZyme	vanilla	84.5 $\pm$ 0.1	68.4 $\pm$ 0.0	65.7 $\pm$ 0.0	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.3 $\pm$ 0.0	100.0	0.94
	filt30	85.3 $\pm$ 1.1	68.4 $\pm$ 0.1	65.4 $\pm$ 0.2	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.6 $\pm$ 0.4	99.1	0.88
	filt50	85.1 $\pm$ 1.2	68.4 $\pm$ 0.1	65.4 $\pm$ 0.3	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.6 $\pm$ 0.4	99.1	0.88
	filt70	85.3 $\pm$ 1.6	68.4 $\pm$ 0.1	65.4 $\pm$ 0.2	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.6 $\pm$ 0.6	100.0	0.87
EnzBERT	vanilla	85.7 $\pm$ 1.9	68.4 $\pm$ 0.1	65.3 $\pm$ 0.2	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.8 $\pm$ 0.7	100.0	0.84
	filt30	85.5 $\pm$ 2.0	68.4 $\pm$ 0.1	65.4 $\pm$ 0.2	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.7 $\pm$ 0.8	99.1	0.84
	filt50	85.4 $\pm$ 1.9	68.4 $\pm$ 0.1	65.4 $\pm$ 0.2	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.7 $\pm$ 0.7	100.0	0.86
	filt70	85.1 $\pm$ 1.8	68.4 $\pm$ 0.1	65.4 $\pm$ 0.2	40.0 $\pm$ 0.0	81.8 $\pm$ 0.0	78.6 $\pm$ 0.7	98.1	0.87

S7.3 Computational cost

We measured aggregation and MILP-construction time while subsampling the SEED universal from 10^3 to 47,880 reactions. Aggregation increases from 0.02 to 0.29 s and construction from 0.06 to 5.9 s. Solve time dominates because the optimiser must choose a biomass-feasible candidate subnetwork from the universal.

On the 108-genome panel at the configuration this paper reports (DeepProZyme-vanilla, evidence-weighted cost, $\delta=0$), the MILP solve call takes a median of 137 s per genome and a mean of 211 s, ranging from 72 to 738 s, on four CPU cores of an Intel Xeon Gold 6242. These are the wall times recorded by the solver-status run in Section S1.1. Table S1 adds the median for the other eleven configurations (226–716 s). Across the 108 genomes solve time is weakly negatively correlated with candidate-set size (Spearman $\rho = -0.34$, $p = 3 \times 10^{-4}$) and, on the re-solved timings of Table S7, with the number of evidence reactions ($\rho = -0.39$): genomes with less evidence force the solver to assemble biomass from expensive scaffold reactions. The candidate range on this panel is narrow (16,018–17,879), so this does not measure scaling to larger universals.

For scale against another tool we have only the earlier measurement: across the 109-genome panel and all twelve baseline $\times$ cutoff configurations, METEOR’s median per-genome build (solve, submodel construction, and MEMOTE evaluation together) was 6.7 minutes on four cores against CarveMe’s 20.3 minutes on two, so roughly three times faster in wall-clock time and about $1.5\times$ cheaper in core time. That comparison predates the move to the 108-genome panel and we have not repeated it; we report it as an order-of-magnitude statement rather than a matched benchmark.

The reported METEOR run uses the open-source CBC solver. Per-protein EC inference is a separate GPU pre-step and is not included in these timings. METEOR parallelises across genomes via a SLURM job array.

S7.4 The same-predictor threshold-and-repair baseline

Main-text Section 3.4 and Section S7.5 compare METEOR against a baseline built from the same predictor. The comparison holds the score source and biochemical database fixed, but the methods use different selection objectives and candidate masks. We give the construction here because several results depend on it.

Draft. For each EC we take the maximum score over the proteome (DeepProZyme, vanilla). A reaction enters the draft if any EC annotated to it reaches $\tau=0.5$. This is the hard-threshold reading of the same scores METEOR consumes.

Gap-fill. The draft alone does not grow (main-text Section 3.1: 0/108 for every predictor), so we complete it with the parsimony repair METEOR uses internally: a linear program over the reactions that can carry flux under the Gram-specific bounds, which holds $\mathbf{S}\mathbf{v} = \mathbf{0}$ and $v_{\text{biomass}} \geq 0.1$ while minimising flux through reactions outside the draft. The draft is therefore reused at no cost and only bridging reactions are added. Should that prove infeasible, we retry over the unrestricted universal.

Shared components and remaining differences. Both arms use the SEED universal, Gram-specific tight bounds, defined medium, biomass floor $\gamma_{\min} = 0.1$, and CBC. The baseline’s repair

LP can draw from all non-excluded reactions and minimises flux outside the thresholded draft. METEOR restricts candidates using $w_j \geq 0.01$, the medium, and the shared biomass skeleton, then charges evidence-derived cost once per selected binary reaction. The comparison therefore evaluates the complete two-stage baseline against the complete METEOR selection procedure; it does not isolate a single algorithmic component.

Which METEOR output is scored. This comparison and Section S7.5 score the ECs of reactions with $y_j > 0.5$ (1,402 per model on the six curated organisms). Section S7.1 (Table S16) and main-text Section 3.4 instead score the ECs in the full reaction-associated EC set (1,521 per model). The two sets come from the same selected reactions; the restricted set keeps only ECs present in the predictor’s vocabulary, whereas the full set keeps every EC number SEED lists for those reactions, including superseded numbers that the vocabulary no longer contains but the curated models still use (503 of the 509 four-digit SEED EC numbers absent from the vocabulary are transferred numbers, the other six being SEED placeholders). The reaction-mapped set is therefore a strict subset in all six organisms (108–124 ECs differ). Against the identical reference sets (530–870 ECs), recall is 0.884 on the full reaction-associated set and 0.784 on the vocabulary-restricted set. The separate protein score matrix E^{new} retains the input columns; ECs outside that vocabulary have organism-level support only.

Across the 25 genomes on which the two arms were compared, the gap-fill adds a mean of 85 reactions (range 54–253) and both arms grow in all 25. The EC-level recall figures quoted in main-text Section 3.4 come from the six of those genomes with a published curated model.

This baseline is distinct from the COBRApy gap-filler timing comparison in Section S4: it uses METEOR’s repair LP, and all resulting models grow.

S7.5 Curated sub-threshold EC recovery

This section supports the main-text result of Section 3.2 with per-organism counts, denominators and interval estimates. Its reference is a published curated model, so unlike the pathway analysis it does not depend on how a pathway is defined or on how many ECs a method returns.

For each genome let W be the set of ECs that the curated model contains and that the predictor scores in $(0, 0.5)$. These ECs are present in the curated model but invisible to a hard threshold. Averaged over the six genomes $|W| = 102$ ECs. We ask what fraction of W each method ends up including.

Output size and the recovery difference. main-text Section 3.2 shows that a coverage advantage can disappear once the baseline is matched in size, so the same confound has to be addressed here. The two arms differ modestly in output size: METEOR emits a mean of 1,402 ECs against the baseline’s 1,274, 10% more, while recall on the sub-threshold set rises by 26% in relative terms. The additional calls concentrate on that set. Across the six models METEOR emits 773 more ECs than the baseline and 77 of them are correct against the curated reference, of which 75 are curated sub-threshold ECs; the remaining 696 are false positives, so the marginal precision of the extra output is 0.10. Table S18 runs the size-matched control of main-text Section 3.2 on this comparison: for each organism the baseline is the K SEED-

representable ECs with the highest maximum DeepProZyme score, K being METEOR’s EC count. METEOR recovers more of W than the size-matched baseline in all six organisms (mean 0.606 against 0.483) at higher precision (0.422 against 0.385) and higher recall of the full reference (0.784 against 0.715). The top- K set includes 423–510 ECs scored below the 0.5 threshold, so it is not a thresholded draft; it is the same number of ECs chosen by score alone, and score alone recovers fewer of the curated sub-threshold functions than growth-constrained selection. This control run counts $|W| = 122$ for *S. aureus* against 123 in Table S19, one EC apart; METEOR’s mean recovery is correspondingly 0.606 here against 0.604 there. Main-text Section 3.2 reports 0.604, the $n=612$ ($|W| = 123$ for *S. aureus*) figure.

Table S18. Size-matched control for curated sub-threshold EC recovery. K : METEOR’s reaction-mapped EC count; top- K : the K highest-scoring SEED-representable ECs. Recovery is the fraction of W included; precision is against the full curated EC set.

Organism	K	$ W $	METEOR	top- K	Prec. METEOR	Prec. top- K
<i>E. coli</i>	1,491	93	0.656	0.538	0.474	0.432
<i>Salmonella</i>	1,459	84	0.560	0.429	0.471	0.433
<i>K. pneumoniae</i>	1,549	90	0.633	0.489	0.462	0.425
<i>P. putida</i>	1,502	119	0.613	0.555	0.419	0.390
<i>S. aureus</i>	1,104	122	0.525	0.426	0.339	0.305
<i>B. subtilis</i>	1,309	102	0.647	0.461	0.369	0.326
Mean		102	0.606	0.483	0.422	0.385

Table S19. Recovery of curated-reference sub-threshold ECs, DeepProZyme scores, six curated GEMs. Recovery is the fraction of W the method includes; precision is against the full curated EC set. Per-organism and pooled intervals are Wilson 95% intervals on the recovered proportion; the mean interval is a bootstrap over organisms. The baseline is the same-predictor threshold-and-repair arm reported in main-text Section 3.2.

Organism	$ W $	baseline	METEOR	METEOR 95% CI
<i>E. coli</i>	93	0.495	0.656	[0.555, 0.745]
<i>Salmonella</i>	84	0.452	0.560	[0.453, 0.661]
<i>K. pneumoniae</i>	90	0.478	0.633	[0.530, 0.726]
<i>P. putida</i>	119	0.479	0.613	[0.524, 0.696]
<i>S. aureus</i>	123	0.431	0.520	[0.433, 0.607]
<i>B. subtilis</i>	103	0.544	0.641	[0.545, 0.727]
Mean recovery	102	0.480	0.604	[0.562, 0.640] [†]
Pooled recovery	612	0.479	0.601	[0.562, 0.639]
Mean precision	–	0.455	0.422	–

[†]Bootstrap over the six organisms, 10^4 resamples, seed 0; the pooled row uses a Wilson interval on 368/612. Baseline pooled recovery is 293/612.

METEOR recovers 0.604 of W against the baseline’s 0.480 (Table S19), higher in all six genomes ($p=0.031$, the smallest value six paired samples admit). Precision falls by 0.032, also in all six. The threshold discards every sub-threshold EC from the draft; METEOR’s complete selection procedure includes a further twelve percentage points of them and pays for it in precision. Because the objectives and candidate masks differ, this comparison does not attribute the difference to the cost term alone.

Recovered functions. The ECs METEOR recovers and the baseline misses include specific metabolic capabilities as well as questionable matches, illustrating both the signal and the metric’s limits. We checked every example below against the ENZYME¹¹ flat files (<https://enzyme.expasy.org/EC/>) and KEGG (<https://rest.kegg.jp/get/ec:>); all are current EC numbers, none transferred or withdrawn. Counts are the organisms in which the EC is recovered by METEOR only, out of those in which it is a curated sub-threshold EC.

- EC 1.1.1.133, dTDP-4-dehydrorhamnose reductase (2/2): supplies dTDP-L-rhamnose for O-antigen assembly. The official equation is written in the oxidative direction although the enzyme operates as a reductase within the dTDP-L-rhamnose synthase complex.
- EC 1.17.1.4, xanthine dehydrogenase (1/1): purine catabolism; distinct from EC 1.17.3.2 (xanthine oxidase).
- EC 1.2.1.21, glycolaldehyde dehydrogenase (2/6): its sole cross-referenced protein in ENZYME is P25553 (ALDA_ECOLI), the *E. coli* enzyme.
- EC 1.3.8.1, short-chain acyl-CoA dehydrogenase (3/4) and EC 1.3.3.4, protoporphyrinogen oxidase (2/3): fatty-acid β -oxidation and haem biosynthesis.

One recovery is worth flagging against ourselves: EC 2.1.1.37, DNA (cytosine-5)-methyltransferase, recovered in *B. subtilis*, is a DNA-modification rather than an intermediary-metabolism enzyme, reaching KEGG metabolism through its S-adenosyl-L-methionine cofactor. It is in the curated model’s EC annotation and so counts as a recovery under this metric, which illustrates the metric’s limit: it scores EC-set membership, not contribution to the network.

Two limits apply to this result. Six genomes cannot establish more than a direction, and it was measured on DeepProZyme alone, so we do not know whether the margin holds for predictors whose score distributions differ (main-text Section 3.1 shows those distributions vary widely).

S7.6 Gene essentiality

Gene–reaction associations are read off the protein-by-EC evidence matrix (each protein maps to one gene locus; single-gene deletion plus FBA on the selected network gives a predicted essential/non-essential call). We tested this against experimentally determined essential-gene sets on six organisms under eight organism-medium-reference combinations (Table S20), each scored against the curated-model ceiling that reference supports (Table S21).

Combination 5’s mean hides high per-predictor variance in opposite directions (CLEAN

Table S20. Gene essentiality, mean MCC over three predictors (CLEAN, DeepProZyme, EnzBERT), METEOR against a threshold-and-repair baseline.

#	Organism	Medium	Reference	METEOR	Threshold
1	<i>E. coli</i>	Minimal (M9)	gess-ecol ¹²	0.302	0.300
2	<i>B. subtilis</i>	Rich	gess-bsub ¹³	0.323	0.247
3	<i>E. coli</i>	Rich	PEC ¹⁴	0.064	0.127
4	<i>B. subtilis</i>	Minimal (M9)	Koo et al. ¹⁵ (assembled)	0.287	0.230
5	<i>S. aureus</i>	Rich	Koo et al. ¹⁵ (cross-species)	0.053	0.053
6	<i>Salmonella</i>	Rich	TraDIS ¹⁶	0.073	0.140
7	<i>K. pneumoniae</i>	Rich	TraDIS ¹⁶	0.086	0.156
8	<i>P. putida</i>	Rich	Tn-seq ¹⁷	0.073	0.122

0.150 vs. 0.065; DeepProZyme -0.010 vs. 0.124 ; EnzBERT 0.020 vs. -0.029 ; METEOR then threshold), against a reference denominator of only 152–200 genes per predictor and a low curated ceiling (Table S21); we report it as an inconclusive tie, not a stable result. Combinations 6–8 were solved only under the rich medium: the essentiality references available for these strains (Ghomi et al. 2024, TraDIS on LB agar; Royet et al. 2025, Tn-seq screened primarily for metal tolerance with an LB baseline) do not include a minimal-medium condition, and we could not locate one in the literature for these strains, so METEOR was not solved under a minimal medium for them either.

Table S21. Curated-model ceiling for each reference table (same single-gene deletion procedure applied to the curated model itself).

Curated model	Reference	MCC
iML1515 (<i>E. coli</i>)	gess-ecol	0.758
iML1515	PEC	0.592
iYO844 (<i>B. subtilis</i>)	gess-bsub	0.390
iYO844	Koo et al. (assembled)	0.666
iYS854 (<i>S. aureus</i>)	Koo et al. (cross-species)	0.304
STM_v1_0 (<i>Salmonella</i>)	TraDIS	0.362
iYL1228 (<i>K. pneumoniae</i>)	TraDIS	0.301
iJN1463 (<i>P. putida</i>)	Tn-seq	0.351

E. coli: reference table, not medium, drives the reversal. A 2×2 decomposition (medium solved under $\times$ reference scored against) separates the two factors (Table S22). Switching the reference table flips a tie into a $2 \times$ threshold advantage; switching medium alone changes scores by only 0.01–0.05. This tracks a real biological effect, medium-dependent essentiality in *E. coli* (~ 119 genes, $\sim 3\%$ of the genome, essential in minimal but not rich medium¹⁸), expressed through which reference table is used rather than through METEOR’s own response to the medium it is solved under.

The same decomposition on *B. subtilis* (Table S23) shows a different pattern: METEOR

Table S22. *E. coli* 2×2: METEOR / threshold mean MCC.

Medium \ Reference	gess-ecol	PEC
Minimal (M9)	0.302 / 0.300	0.052 / 0.105
Rich	0.248 / 0.266	0.064 / 0.127

leads in all four cells, and although the reference-table effect is again larger than the medium effect (largest single-cell change -0.068), it never reverses the ranking. The asymmetry is itself the finding: how sensitive METEOR’s advantage is to reference-table choice varies by organism and is not a general property of the method.

Table S23. *B. subtilis* 2×2: METEOR / threshold mean MCC.

Medium \ Reference	gess-bsub	Koo et al.
Rich (LB)	0.323 / 0.247	0.255 / 0.222
Minimal (M9)	0.325 / 0.239	0.287 / 0.230

E. coli: gene-level effect of the medium switch. For each of the 3,888 genes common to the model and both reference tables, we compared the essential/non-essential call under the minimal and rich media (Table S24), against the 119 genes Joyce et al.¹⁸ report as genuinely medium-dependent. METEOR flips a call on medium change about twice as often as the threshold baseline (mean 1.72% vs. 0.93% of genes) and recovers more of the true Joyce flips in the correct direction (30 of 37 flips attempted vs. 25 of 26). Where METEOR flips in the wrong direction, the errors are not scattered: under DeepProZyme all five flip errors are the de novo purine-synthesis genes *purC*, *purD*, *purF*, *purL* and *purM* together, and under EnzBERT both errors are the branched-chain amino-acid genes *ilvC* and *ilvD* together. The threshold baseline attempted far fewer flips and made one error (a single isolated gene).

Table S24. Gene-level flip rate on the minimal/rich medium switch (*E. coli*, $n = 3,888$ common genes). Correct/wrong direction are counted only among the 119 genes Joyce et al. mark as truly medium-dependent.

	Flip rate (mean)	Correct direction	Wrong direction
METEOR (3 predictors)	1.72%	30	7
Threshold (3 predictors)	0.93%	25	1
Real biology (Joyce et al.)	3.06% (119/3,888)	–	–

Salmonella, *K. pneumoniae* and *P. putida*: candidate-pool contamination, and METEOR’s exclusions checked against curated models. We traced the threshold baseline’s advantage on these three organisms (all nine organism×predictor cells) to two largely independent sources.

First, the shared candidate pool contains reactions annotated as eukaryotic that the threshold baseline’s gap-fill admits regardless of evidence; a mitochondrial deoxynucleotide (dATP) carrier appears in the threshold-only reaction set in all nine cells, and a second (dCTP) carrier in six of nine. Excluding reactions carrying a eukaryotic-organelle annotation and re-solving all nine cells reduces the mean threshold-minus-METEOR MCC gap from 0.062 to 0.031 and reverses the ranking in one cell (*P. putida*, CLEAN: -0.032 , METEOR ahead). Second, of the reactions METEOR still declines that have no alternative route in its own selected network (evidence ≥ 0.5 but not selected, and no same-EC substitute or non-cofactor metabolite connectivity to the selected network; 95 such reactions across the nine cells), only 19% are found in that organism’s curated model by exact ModelSEED reaction ID or EC-code cross-reference, against 74% for the reactions METEOR keeps under the same criteria ($n = 25,332$). METEOR’s exclusions therefore agree with curated annotation about as strongly as its inclusions do, so most of the residual gap on these three organisms reflects candidate-pool quality rather than a weakness in which reactions METEOR chooses to keep or drop.

Koo et al. reference construction. The *B. subtilis* minimal-medium reference (combination 4) is assembled, not a single deposited table: its positive class (370 genes) is the union of Koo et al.’s supplementary Table S3 (257 genes essential in rich medium) and their Table S4 (98 nutrient auxotrophs identified in minimal medium); the two do not overlap, since their Table S3 genes are not eligible for their Table S4 knockout screen. Koo et al. report no non-essential class, so the negative class (667 genes) is drawn from gess-bsub, excluding genes already in the 370-gene union (170 genes) and genes gess-bsub calls essential that Koo et al. do not confirm (7 genes, discarded as unresolved rather than assigned either way). The *S. aureus* reference (combination 5) is a different Koo et al. table: a 392-gene cross-species comparison (*B. subtilis*/*S. sanguinis*/*S. aureus*/*E. coli*) sourced from Chaudhuri et al. 2009 and Santiago et al. 2015.

S7.7 Per-predictor structural properties

Main-text Figure 2a reports dead-end metabolites per genome and growth outcome for all three predictors; Table S25 adds the two per-predictor structural metrics the main text otherwise states only as aggregate ranges (main-text Section 3.1): selected-reaction counts and the mass-imbalanced fraction. The mass-imbalanced range quoted in the main text (0.49–0.51 for baseline, 0.46–0.50 for METEOR) is the minimum and maximum of this table’s six values; no additional claim in the paper depends on the per-predictor breakdown itself.

S7.8 Recovery stratified by predictor vocabulary

For each predictor and each of the six curated organisms we split the curated reference ECs by whether the predictor’s score matrix has a column for them (in or out of vocabulary) and report the fraction of each group recovered by three arms: the $\tau=0.5$ threshold, the skeleton-only selection of Table S7, and METEOR’s full reaction-associated EC set. EC numbers in all sets are mapped to their current versions with the same obsolete-to-current table before comparison, which raises DeepProZyme’s overall METEOR recall slightly (0.890, against 0.884

Table S25. Structural properties of metabolic networks across the 108-GCF panel (mean $\pm$ s.d. over genomes). Baseline networks are per-EC threshold networks: they contain every reaction for which at least one protein scores ≥ 0.5 on one of its ECs; METEOR networks contain the reactions selected using the corresponding predictor. Dead-end and mass-imbalanced counts follow the definitions in main-text Section 2.2.

Metric	CLEAN	DeepProZyme	EnzBERT	+CLEAN	+DeepProZyme	+EnzBERT
	Baseline $\tau = 0.5$			METEOR		
Selected reactions	2,003 $\pm$ 614	2,810 $\pm$ 780	2,225 $\pm$ 620	1,890 $\pm$ 531	3,121 $\pm$ 786	2,228 $\pm$ 595
Dead-end metabolites $\downarrow$	7.4 $\pm$ 2.3	9.8 $\pm$ 4.6	5.7 $\pm$ 3.8	0.7 $\pm$ 0.6	2.4 $\pm$ 1.0	1.0 $\pm$ 0.6
Mass-imbalanced fraction $\downarrow$	0.508 $\pm$ 0.031	0.489 $\pm$ 0.029	0.502 $\pm$ 0.028	0.501 $\pm$ 0.023	0.464 $\pm$ 0.027	0.488 $\pm$ 0.025
Growth-feasible genomes $\uparrow$	0/108	0/108	0/108	108/108	108/108	108/108

Note. $n=108$ genomes per column; per-genome distributions in main-text Figure 2a.

Growth is FBA on the induced submodel, no gap-filling, every exchange open, both arms alike.

in Table S16 without the mapping). A threshold cannot recover an out-of-vocabulary EC by construction; the skeleton recovers 31–40% of them, and METEOR 55–71%, higher than the skeleton alone in all six organisms for each predictor (Wilcoxon $p=0.031$, 6/6). The remaining ceiling is set by the candidate set: 2.5–4.4% of curated ECs map to no candidate reaction.

Table S26. Curated-EC recall stratified by predictor vocabulary, six curated organisms (mean $\pm$ s.d.). Out: curated ECs absent from the predictor’s vocabulary; unreach.: curated ECs with no candidate reaction.

Predictor	Out	Out of vocabulary			In vocabulary			Unreach.
		Thr.	Skel.	METEOR	Thr.	Skel.	METEOR	
CLEAN	6.9%	0	0.31 $\pm$ 0.01	0.66 $\pm$ 0.06	0.72 $\pm$ 0.11	0.58 $\pm$ 0.03	0.84 $\pm$ 0.07	4.2%
DeepProZyme	12.4%	0	0.33 $\pm$ 0.05	0.71 $\pm$ 0.04	0.81 $\pm$ 0.07	0.59 $\pm$ 0.02	0.92 $\pm$ 0.03	2.5%
EnzBERT	10.3%	0	0.40 $\pm$ 0.02	0.55 $\pm$ 0.07	0.70 $\pm$ 0.06	0.58 $\pm$ 0.03	0.84 $\pm$ 0.03	4.4%

S8. Full-proteome protein-level comparison

To examine protein-level fidelity beyond the 66-protein holdout set, we evaluated METEOR on all proteins carrying four-digit EC annotations in the NCBI RefSeq records (GenBank flat file format) of the 95 genome-collection assemblies whose RefSeq records include such annotations (94 for the CLEAN-filt30 configuration). For each genome, we downloaded the GBFF file via the NCBI Datasets API, extracted EC-number qualifiers from CDS features, and matched proteins by their RefSeq accessions (NP_/WP_/YP_) to the baseline and METEOR prediction tables.

Scope and limitation. These NCBI-annotated proteins include baseline training examples. The comparison describes agreement with RefSeq EC annotations, including computational annotations, on a shared input set. METEOR additionally uses reaction mappings and metabolic constraints. An independent test of annotation accuracy requires experimentally supported, held-out labels.

Under the fair net-only comparison, the baseline is mapped through the same EC vocabulary as METEOR, so that the vocabulary normalisation is not mis-credited to METEOR. The

score feedback does not improve top-1 accuracy on the full proteome. For DeepProZyme-vanilla ($n=69,666$), top-1 is $90.2\% \rightarrow 90.1\%$ (103 corrections vs. 188 regressions) and $F_{\max}$ is unchanged; the only positive effect is a small decision-threshold recovery (micro- $F_1@0.5 +0.002$). This is consistent with the holdout ($F_1@0.5 +0.008$; top-1 unchanged and $F_{\max}$ nearly unchanged) and shows little change in per-protein performance.

S8.1 Net-only correction rate by training-set sequence identity

Stratifying the net-only top-1 outcome by each protein’s maximum identity to the DeepProZyme training set (DIAMOND¹⁹) shows the per-protein effect is positive by three proteins in the lowest-identity bin and negative in every other bin (Table S27). This descriptive difference does not establish that the metabolic constraint helps low-identity proteins. The inflated correction counts of earlier non-net-only comparisons were an artefact of an unnormalised baseline vocabulary.

Table S27. Net-only top-1 corrections/regressions by maximum training-set sequence identity (DeepProZyme, $n=69,666$). Net = corrections – regressions.

Identity	n	top-1 B	top-1 $+M$	Corr	Regr
<30%	2,911	73.8%	73.9%	21	18
30–50%	17,013	87.4%	87.3%	38	64
50–70%	15,838	93.2%	93.0%	8	28
70–90%	11,234	95.1%	95.0%	1	11
>90%	19,801	95.5%	95.4%	6	19
no hit	2,869	52.0%	51.4%	29	48

S9. Internal deletion-recovery stability

Setup as in main-text Section 3.3 (the same $D=60$ deletion, four cost regimes, on the full 108-genome panel); Table S28 adds the Gram-type split.

Table S28. Internal deletion-recovery stability on the full 108-genome panel (mean off-reference reactions per genome), overall and split by Gram type.

Regime	all ($n=108$)	Gram– ($n=63$)	Gram+ ($n=45$)
evidence-derived	34.6	31.2	39.4
two-stage	78.2	70.1	89.6
uniform	142.9	133.8	155.6
shuffled	2,006.6	1,958.7	2,073.6

Recovery precision, the fraction of newly selected reactions that belong to the sampled set D , is low throughout: 0.037 for the evidence-derived cost, 0.022 for two-stage, 0.008 for

uniform, and 0.004 for shuffled costs (the two-stage arm’s objective-shape caveat is stated in main-text Section 3.3).

This experiment measures stability relative to METEOR’s own output, not biological accuracy. It uses one fixed sample per genome, and the deposited records do not retain the identities of the sampled reactions.

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